# $H_2SO_4$

## Properties:

M.W = 98.08 g/mol

Boiling point = 10.5°C

Melting point = 340°C

## Solubility:

Soluble in water → produces large amount of heat

→ $SO_3$ soluble in $H_2SO_4$

## Production of $H_2SO_4$:

## Contact Process:

I. $S_{(s)} + O_2 \longrightarrow SO_2$
(S = elemental state)

II. $\boxed{SO_2 + \tfrac{1}{2}O_2 \xrightleftharpoons{450^\circ C} SO_3}$ RDS

III. $SO_3 + H_2SO_4 = H_2S_2O_7$ (Oleum)

IV. $H_2S_2O_7 + H_2O = 2H_2SO_4$

Catalyst: $V_2O_5$/Suitable support

## Catalyst for the Oxidation of $SO_2$ to $SO_3$

| | Current | Previously used |
|---|---|---|
| a. | $V_2O_5/SiO_2$ | $Pt/SiO_2$ |
| b. | $V_2O_5/Al_2O_3$ | $Pt/Al_2O_3$ |
| c. | $V_2O_5/SiO_2-Al_2O_3$ | $Pt/SiO_2-Al_2O_3$ |
| d. | $V_2O_5$/Kheiselguhr | Pt/Kieselguhr |
| 99% conv. | (7–8%) $V_2O_5$ loading | < 0.5 wt%. Pt loading |

**Disadvantages of Pt based Catalyst:**

(a) Poisoning effect

(b) Rapid heat deactivation

(c) Initial ~~d.~~ increment is high.

(d) fragile

Feed: Concentrated ($SO_2$, $O_2$)

$NH_4VO_3$ → Ammonium Vanadate

**Advantages of $V_2O_5$ based Catalyst:**

a. Cost effective

b. Immune to poisoning

c. Stable at 450°C.

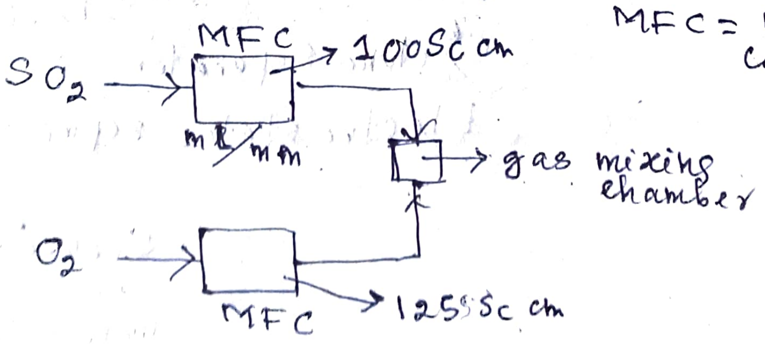
**disadvantage of $V_2O_5$ based Catalyst:**

$[SO_2]$ = 7–10%

$1\% = 10^4$ ppm

$[O_2]$ = 11–14%

MFC = mass flow configuration.



$NH_4VO_3$ ($V_2O_5$) + $H_2O$ $\longrightarrow$ $NH_4VO_3$ (aq)

<u>Basis</u> $\rightarrow$ 1g of $SiO_2$ support

$\downarrow$ thoroughly mixed

$NH_4VO_3/SiO_2$ (Paste)

$\downarrow$ Muffle furnace 550°C, Air atm 2h

$V_2O_5/SiO_2$ (lump)

$\downarrow$ Ball milling

$V_2O_5/SiO_2$ (powder)

$\downarrow$ hydrolic press

$V_2O_5/SiO_2$ pallete (3mm diameter)

<u>Probable sources of $SO_2$ :</u>

I. $H_2S$ + $\longrightarrow$ $SO_2$ (KLAUS Process - 1st step)

II. Spent $H_2SO_4/FeSO_4$ (steel industries)

$\downarrow \Delta$

$SO_2$ (Dust particle, Acidic impurities (As, F, Cl))

Cyclone Separator

Electrostatic separator

Scrubbing with monoethanol amine

III. Smelter sources: (non-ferrous material)

- $CuS$, $PbS$, $ZnS$, $MoS_2$

Molybdenum disulphide.

$$\downarrow +O_2$$

$SO_2$ (impure)

$\downarrow$ Purified

IV. $FeS_2$ (40-45% S)

S, As,

$\downarrow$ Finnish Process (1300°C)

$SO_2$ (impure) (1000°C-1100°C)

$\downarrow$

Cooling

$\downarrow$

Purification

V. Elemental S: Oxidation reaction

# Process flow diagram for the manufacturing $H_2SO_4$:

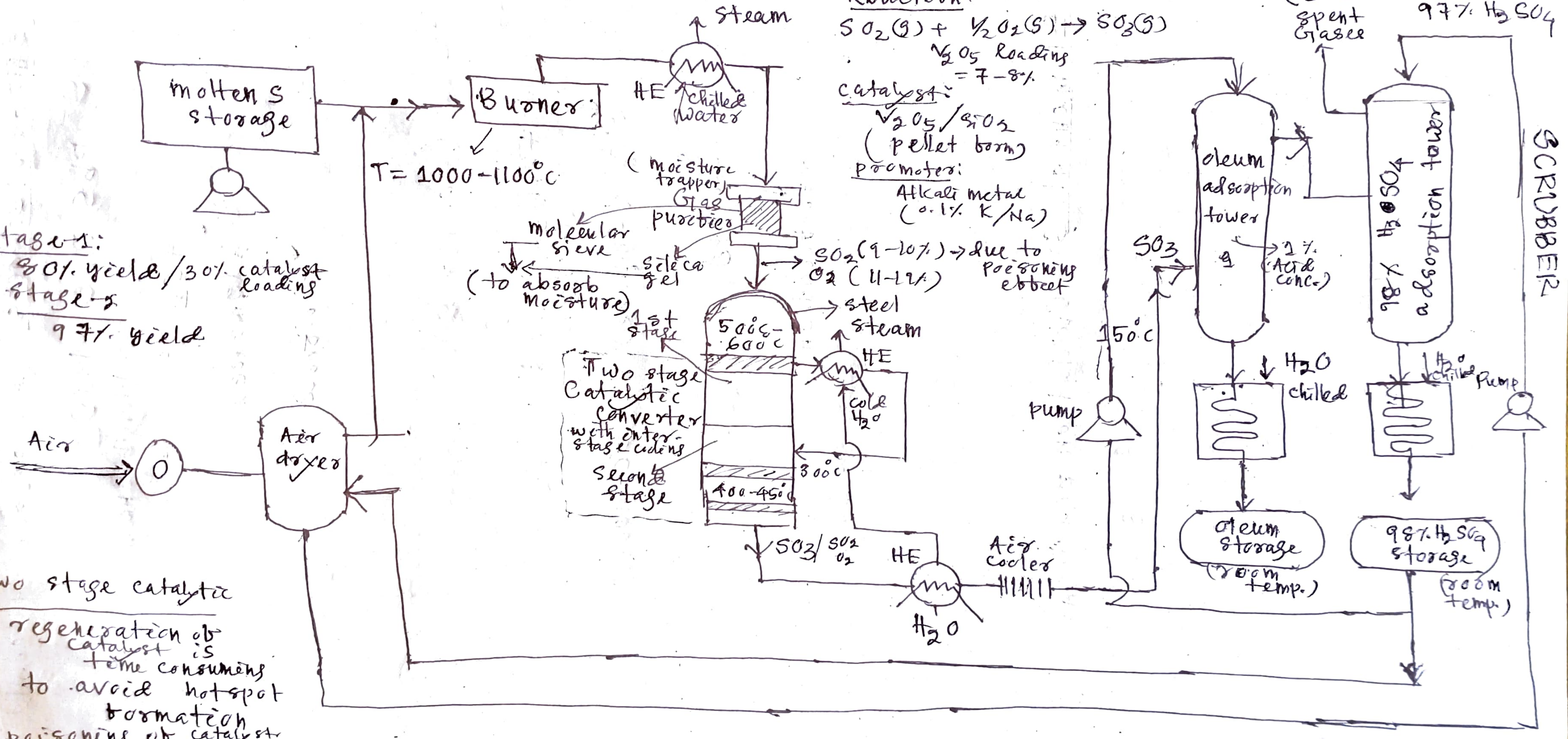
**Reaction:**

$SO_2(g) + \frac{1}{2}O_2(g) \rightarrow SO_3(g)$

$V_2O_5$ loading $= 7-8\%$

**catalyst:** $V_2O_5/SiO_2$ (pellet form)

**promoter:** Alkali metal ($0.1\%$ K/Na)




* **Stage-1:**
80% yield / 30% catalyst loading
**Stage-2**
97% yield

* **Two stage catalytic**
→ regeneration of catalyst is time consuming
→ to avoid hotspot formation
→ poisoning of catalyst

# Polymer fundamentals :

Starting material

↓

monomer

↓ Reactive functional groups | ↓ Double, triple bonds

**Reactive functional groups**

(I) Carboxylic acid ($-COOH$)

(II) Amine group ($-NH_2$)

(III) Amide groups ($-CONH_2$)

(IV) hydroxyl groups ($-OH$)

**Double, triple bonds**

$CH_2{=}CH_2$

functionality → 2

$CH{\equiv}CH$

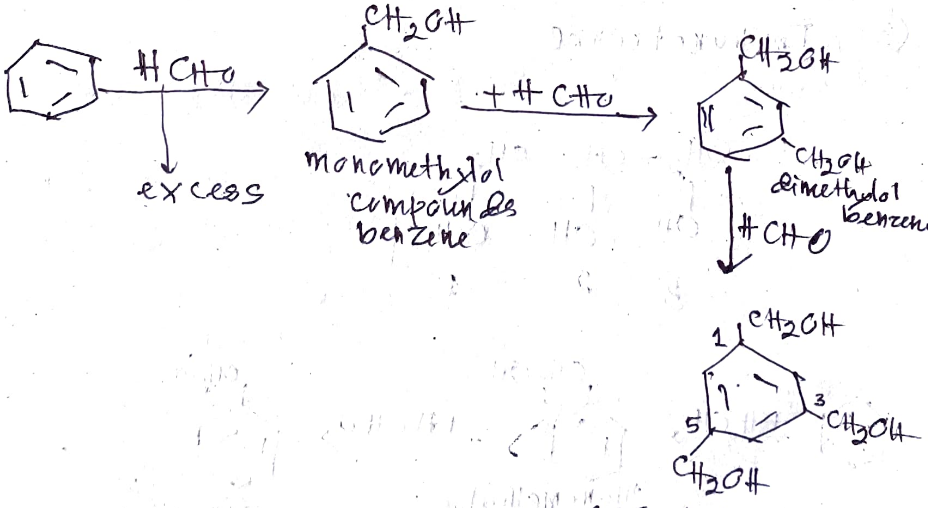
functionality = 4

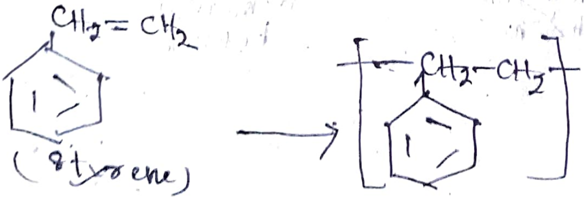
Depending upon the functional groups present –

(I) Unifunctional ($CH_3-COOH$) → Acetic Acid

(II) Bifunctional ($\underset{OH}{CH_2}-\underset{OH}{CH_2}$) → ethylene glycol

(III) tri-functional ($\underset{OH}{CH_2}-\underset{OH}{CH}-\underset{OH}{CH_2}$) → glycerol



Monomeric unit

$\downarrow +M$

Dimer

$\downarrow +M$

Trimer $\xrightarrow{+M}$ Trimer $\xrightarrow{+M}$ Tetramer

$\downarrow +M$

.... Hexamer $\xleftarrow{+M}$ Pentamer

(1) Unifunctional

doesn't take part in polymer formation

inert towards polymer formation.

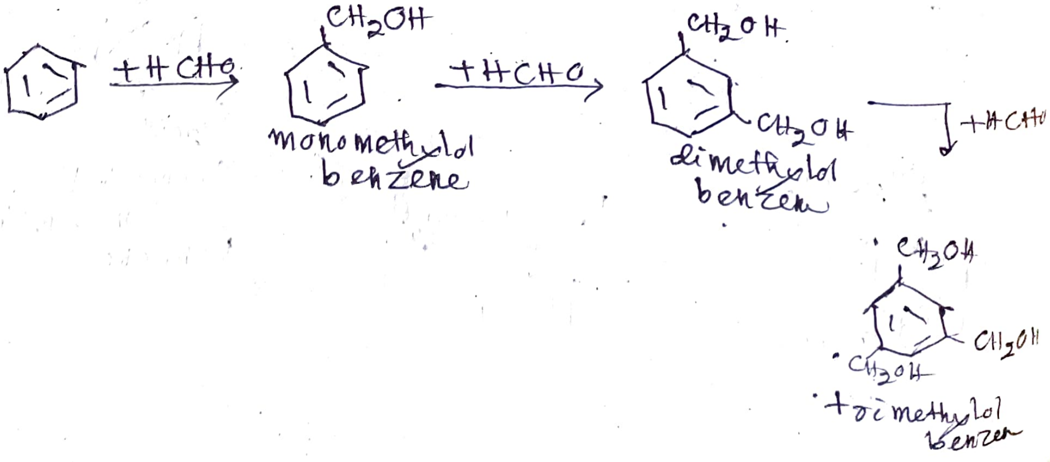
$CH_3-\underbrace{COOH}$ $\rightarrow$ fm = 1

(2) bifunctional

$$\begin{matrix} CH_2 & - & CH_2 \\ | & & | \\ OH & & OH \\ 1 & & 2 \end{matrix}$$

(3) Trifunctional

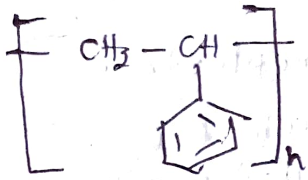
$$\begin{matrix} CH_2 & - & CH & - & CH_2 \\ | & & | & & | \\ OH & & OH & & OH \\ 3 & & 2 & & 1 \end{matrix}$$



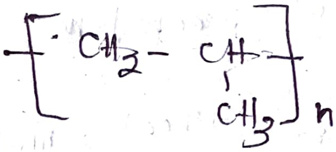
commodity polymers:

ⓐ Polyethylene $\left[ CH_2 - CH_2 \right]_n$

ⓑ Polystyrene



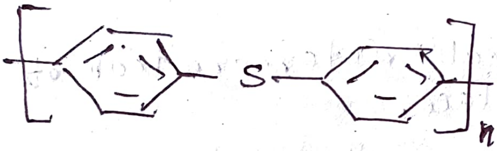
ⓒ Polypropylene



Electronus:

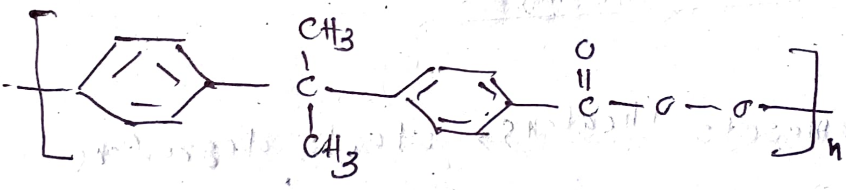
ⓐ Poly acetylene $\left[ CH = CH \right]_n$

ⓑ Polyphynelene sulfide

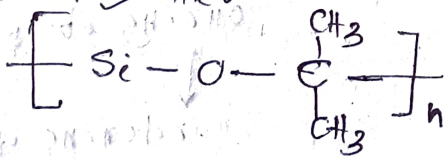


Polymers in biomedical applications:

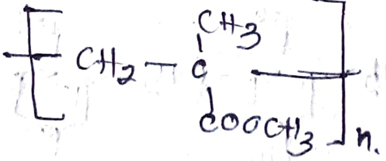
ⓐ Polycarbonate



ⓑ Silicane polymer



ⓒ Polymethyl methacrylate



# Classification of Polymers:

## (a) Based on application of heat

### (i) Thermoplastics:

nylon - 6,6
Alkyl resin
Poly propylene

Thermoplast $\xrightarrow{+\text{Heat}}$ monomer melts
↓ cooling
solidifies

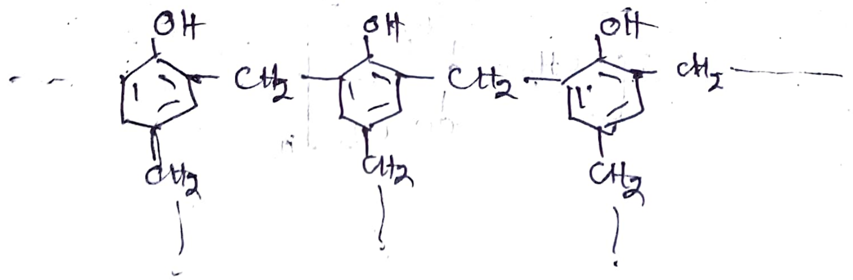
this heating - cooling - heating - cooling cycle is repeated for several times.

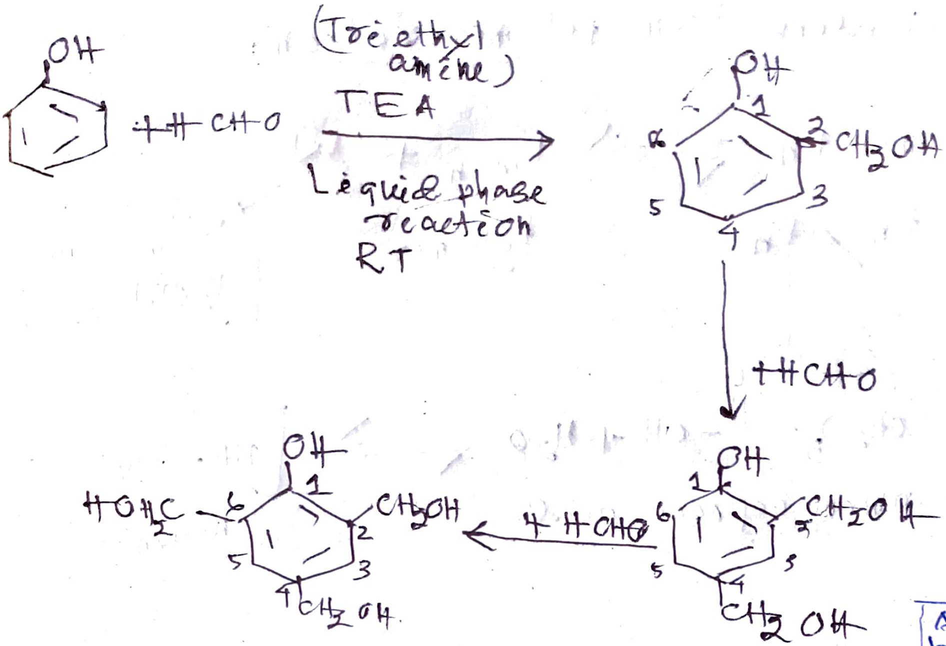
→ no structural deterioration of the polymer takes place.

### (ii) Thermosettings:

Thermosets $\xrightarrow{+\text{heating}}$ starts degrading
↓ cooling
curing of polymer takes place
↓
hardening of polymer

Ex - Bakelite






Starting monomer for Bakelite formation.

Adarsh ko bhejna h
Alkaline m kya hoga.

(b) <u>Based on reaction mechanism.</u>

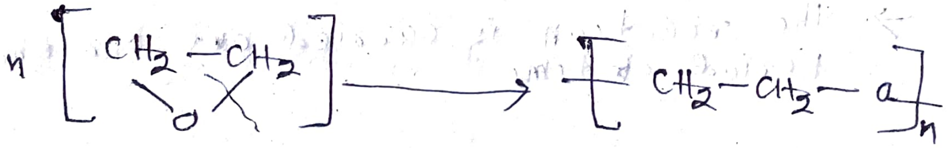
(i) <u>Addition polymerization:</u>

<u>Mechanism:</u>

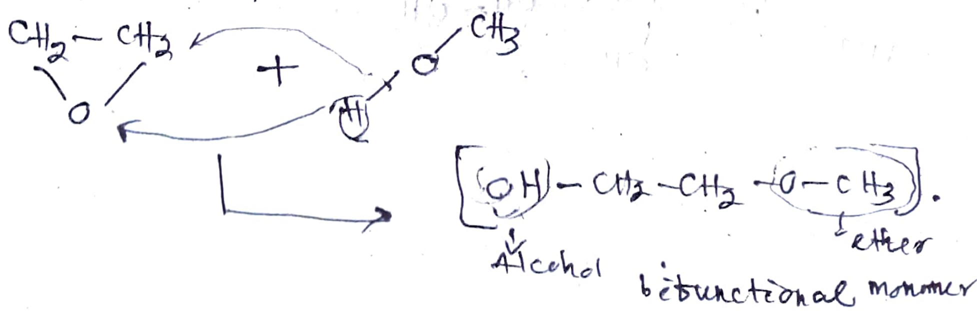
$$\underset{\text{growing chain}}{P_M} \xrightarrow{+M} P_{M+1}$$

(a) $$n[CH_2=CH_2] \xrightarrow[\,[Al(C_2H_5)_3 + TiCl_4]\,]{6-10\,atm\ P,\ 70^\circ C} \left[CH_2-CH_2\right]_n$$

(b) <u>Ring opening reaction</u>




(c)



(ii) Condensation polymerization:

$P_m + P_n \rightleftharpoons P_{m+n} + 'x'$ → small molecules $[H_2O, NH_3, CO_2 \sim]$

$P_m \neq P_n$ Polymer

$NH_2-(CH_2)_5-\overset{O}{\overset{\|}{C}}-OH + H_2O \rightleftharpoons -[NH-(CH_2)_5-\overset{O}{\overset{\|}{C}}]_n- + H_2O$

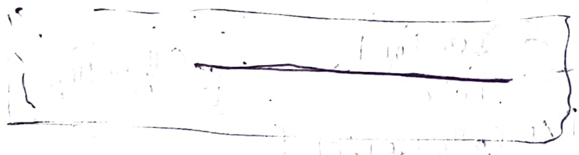
w- amino caproic acid. nylon-6

(c) Based on- molecular structure

(i) Linear chain polymers

$n[CH_2=CH_2] \longrightarrow -[CH_2-CH_2]_n-$

functionality 2

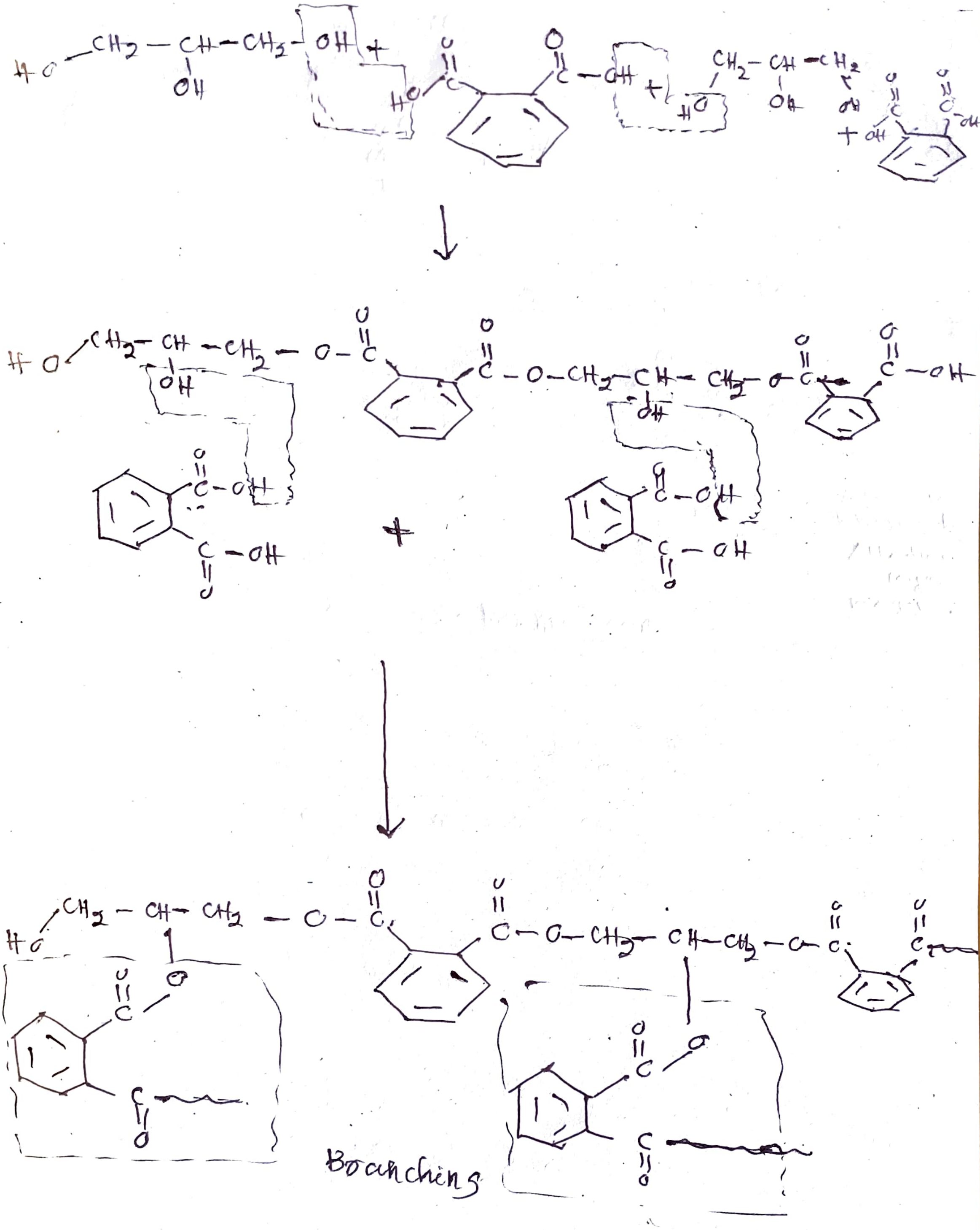


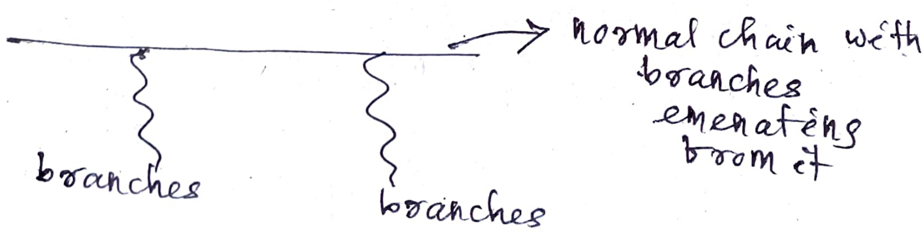
(ii) Branched chain polymer:

→ functionality > 2

→ The reaction is carried out for shorter period of time

$HO-CH_2-\underset{OH}{\underset{|}{CH}}-CH_2-OH$


Branching


normal chain with branches emanating from it
branches
branches